

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication
Kind Code
Publication Date
Inventor(s)

20250277284
A1
September 04, 2025
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A MARTENSITIC STEEL AND A METHOD OF MARTENSITIC STEEL THEREOF

Abstract

A martensitic steel sheet including of the following elements $0.08\% \leq C \leq 0.14\%$; $1.95\% \leq Mn \leq 2.6\%$; $0.1\% \leq Si \leq 0.8\%$; $0.01\% \leq Al \leq 0.1\%$; $0.001\% \leq Ti \leq 0.1\%$; $0.0001\% \leq B \leq 0.05\%$; $0\% \leq S \leq 0.09\%$; $0\% \leq P \leq 0.09\%$; $0\% \leq N \leq 0.09\%$; $0.1\% \leq Cr \leq 1\%$; $0\% \leq Ni \leq 1\%$; $0\% \leq Cu \leq 1\%$; $0\% \leq Mo \leq 0.4\%$; $0\% \leq Nb \leq 0.1\%$; $0\% \leq V \leq 0.1\%$; $0\% \leq Sn \leq 0.1\%$; $0\% \leq Pb \leq 0.1\%$; $0\% \leq Sb \leq 0.1\%$; $0.001\% \leq Ca \leq 0.01\%$; the remainder composition being composed of iron and unavoidable impurities, the microstructure of said steel including, by area percentage, at least 92% of martensite, a cumulated amount of ferrite and bainite from 1% to 8%, and an optional amount of residual austenite from 0% to 2%.

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Family ID: 81975270
Appl. No.: 18/860905
Filed (or PCT Filed): May 19, 2022
PCT No.: PCT/IB2022/054657

Publication Classification

Int. Cl.: C21D9/46 (20060101); C21D1/18 (20060101); C21D6/00 (20060101); C21D8/02 (20060101); C22C38/00 (20060101); C22C38/02 (20060101); C22C38/04 (20060101); C22C38/06 (20060101); C22C38/26 (20060101); C22C38/28 (20060101); C22C38/32 (20060101); C22C38/38 (20060101)

U.S. Cl.:

CPC C21D9/46 (20130101); C21D1/18 (20130101); C21D6/002 (20130101); C21D6/005 (20130101); C21D6/008 (20130101); C21D8/0205 (20130101); C21D8/0226 (20130101); C21D8/0236 (20130101); C21D8/0263 (20130101); C22C38/001 (20130101); C22C38/002 (20130101); C22C38/02 (20130101); C22C38/04 (20130101); C22C38/06 (20130101); C22C38/26 (20130101); C22C38/28 (20130101); C22C38/32 (20130101); C22C38/38 (20130101); C21D2211/002 (20130101); C21D2211/005 (20130101); C21D2211/008 (20130101)

Background/Summary

[0001] The present invention relates to a method of manufacturing of a martensitic steel suitable for automotive industry, yellow goods and green goods. More particularly related to Martensitic steels having tensile strength 1180 MPa or more.

BACKGROUND

[0002] Automotive parts are required to satisfy two inconsistent necessities, e.g. ease of forming and strength but in recent years a third requirement of improvement in fuel consumption is also bestowed upon automobiles in view of global environmental concerns. Thus, now automotive parts must be made of material having high formability in order that to fit in the criteria of ease of fit in the intricate automobile assembly and at same time have to improve strength for vehicle crashworthiness and durability while reducing weight of vehicle to improve fuel efficiency.

SUMMARY OF THE INVENTION

[0003] Therefore, intense Research and development endeavors are put in to reduce the amount of material utilized in car by increasing the strength of material. Conversely, an increase in strength of steel sheets decreases formability, and thus development of materials having both high strength and high formability is necessitated.

[0004] Earlier research and developments in the field of high strength and high formability steel sheets have resulted in several methods for producing high strength and high formability steel sheets, some of which are enumerated herein for conclusive appreciation of the present invention:

[0005] The steel sheet of WO2017/065371 is manufactured through the steps of: rapidly heating a material steel sheet for 3 to 60 seconds to an Ac3 transformation point or higher and maintaining the material steel sheet, the material steel sheet containing 0.08 to 0.30 wt % of C, 0.01 to 2.0 wt % of Si, 0.30 to 3.0 wt % of Mn, 0.05 wt % or less of P and 0.05 wt % or less of S and the remainder being Fe and other unavoidable impurities; rapidly cooling the heated steel sheet to 100° C./s or higher with water or oil; and rapidly tempering to 500° C. to A1 transformation point for 3 to 60 seconds including heating and maintaining time. But the steel of WO2017/065371 do not mention the tensile strength of 1180 MPa or more in both rolling and transverse direction. Further the hole expansion ratio of the steel of WO2017/065371 is not more than 40% even after having a tempered martensite single phase structure.

[0006] WO2010/036028 relates to a hot dip galvanized steel sheet and a manufacturing method thereof. The hot dip galvanize steel sheet includes a

steel sheet including a martensitic structure as a matrix, and a hot dip galvanized layer formed on the steel sheet. The steel sheet includes C of 0.05 wt % to 0.30 wt %, Mn of 0.5 wt % to 3.5 wt %, Si of 0.1 wt % to 0.8 wt %, Al of 0.01 wt % to 1.5 wt %, Cr of 0.01 wt % to 1.5 wt %, Mo of 0.01 wt % to 1.5 wt %, Ti of 0.001 wt % to 0.10 wt %, N of 5 ppm to 120 ppm, B of 3 ppm to 80 ppm, an impurity, and the remainder of Fe. But the steel of WO2010/036028 does not mention hole expansion ratio.

[0007] It is an object of the present invention is to solve these problems by making available cold-rolled martensitic steel sheets that simultaneously have: [0008] an ultimate tensile strength of 1180 MPa or more in both transversal direction as well as rolling direction and preferably more than 1220 MPa in both transversal direction as well as rolling direction, [0009] a yield strength from 900 MPa to 1180 MPa in both transversal direction as well as rolling direction and preferably from 950 MPa to 1180 MPa in both transversal direction as well as rolling direction, [0010] a total elongation of more than 3% in both transversal direction as well as rolling direction, and [0011] a hole expansion ratio of more than 40% and preferably above 50%.

[0012] The present invention provides a martensitic steel sheet having a composition comprising of the following elements, expressed in percentage by weight: [0013] $0.08\% \leq C \leq 0.14\%$; [0014] $1.95\% \leq Mn \leq 2.6\%$; [0015] $0.1\% \leq Si \leq 0.8\%$; [0016] $0.01\% \leq Al \leq 0.1\%$; [0017] $0.001\% \leq Ti \leq 0.1\%$; [0018] $0.0001\% \leq B \leq 0.05\%$; [0019] $0\% \leq S \leq 0.09\%$; [0020] $0\% \leq P \leq 0.09\%$; [0021] $0\% \leq N \leq 0.09\%$; [0022] and can contain one or more of the following optional elements [0023] $0.1\% \leq Cr \leq 1\%$; [0024] $0\% \leq Ni \leq 1\%$; [0025] $0\% \leq Cu \leq 1\%$; [0026] $0\% \leq Mo \leq 0.4\%$; [0027] $0\% \leq Nb \leq 0.1\%$; [0028] $0\% \leq V \leq 0.1\%$; [0029] $0\% \leq Sn \leq 0.1\%$; [0030] $0\% \leq Pb \leq 0.1\%$; [0031] $0\% \leq Sb \leq 0.1\%$; [0032] $0.001\% \leq Ca \leq 0.01\%$; [0033] the remainder composition being composed of iron and unavoidable impurities caused by processing, the microstructure of said steel comprising, by area percentage, at least 92% of martensite, a cumulated amount of ferrite and bainite from 1% to 8%, and an optional amount of residual austenite from 0% to 2%.

[0034] Preferably, such steel can also have a good suitability for forming, for rolling with good weldability and coatability.

[0035] Another object of the present invention is also to make available a method for the manufacturing of these sheets that is compatible with conventional industrial applications while being robust towards manufacturing parameters shifts.

[0036] The above object and other advantages of the present invention will become more apparent by describing in detail the preferred embodiment of the present invention.

Description

DETAILED DESCRIPTION

[0037] The chemical composition of the cold rolled martensitic steel comprises of the following elements:

[0038] Carbon is present in the steel of the present invention is from 0.08% to 0.14%. Carbon is an element necessary for increasing the strength of the Steel of the present invention by producing a low-temperature transformation phases such as Martensite. Therefore, Carbon plays two pivotal roles, one is to increase the strength. But Carbon content less than 0.08% will not be able to impart the tensile strength to the steel of the present invention. On the other hand, at a Carbon content exceeding 0.14%, the steel exhibits poor spot weldability which limits its application for the automotive parts. A preferable content for the present invention may be kept from 0.09% to 0.13 and more preferably from 0.1% to 0.12%.

[0039] Manganese content of the steel of the present invention is from 1.95% to 2.6%. This element is gammaenous. Manganese provides solid solution strengthening and suppresses the ferritic transformation temperature and reduces ferritic transformation rate hence assist in the formation of martensite. An amount of at least 1.95% is required to impart strength as well as to assist the formation of Martensite. But when Manganese content is more than 2.6% it produces adverse effects such as it retards transformation of Austenite to Martensite during cooling after annealing. Manganese content of above 2.6% can get excessively segregated in the steel during solidification and homogeneity inside the material is impaired which can cause surface cracks during a hot working process. The preferred limit for the presence of Manganese is from 2% to 2.5% and more preferably from 2% to 2.4% and even more preferably from 2% to 2.3%.

[0040] Silicon content of the steel of the present invention is from 0.1% to 0.8%. Silicon is an element that contributes to increasing the strength by solid solution strengthening. Silicon is a constituent that can retard the precipitation of carbides during cooling after annealing, therefore, Silicon promotes formation of Martensite. But Silicon is also a ferrite former and also increases the Ac3 transformation point which will push the annealing temperature to higher temperature ranges that is why the content of Silicon is kept at a maximum of 0.8%. Silicon content above 0.8% can also temper embrittlement and in addition silicon also impairs the coatability. The preferred limit for the presence of Silicon is from 0.15% to 0.7% and more preferably from 0.2% and 0.6%.

[0041] The content of the Aluminum is from 0.01% to 0.1%. in the present invention Aluminum removes Oxygen existing in molten steel to prevent Oxygen from forming a gas phase during solidification process. Aluminum also fixes Nitrogen in the steel to form Aluminum nitride to reduce the size of the grains. Higher content of Aluminum, above 0.1%, increases Ac3 point to a high temperature thereby lowering the productivity. The preferred limit for the presence of Aluminium is from 0.01% to 0.05%.

[0042] Titanium is added to the Steel of the present invention from 0.001% to 0.1%. It forms Titanium-nitrides appearing during solidification of the cast product. The amount of Titanium is so limited to 0.1% to avoid the formation of coarse Titanium-nitrides detrimental for formability. In case the Titanium content below 0.001% does not impart any effect on the steel of the present invention.

[0043] Boron is an essential optional element for the steel of the present invention and may be present from 0.0001% to 0.05%. Boron forms boronitrides and impart additional strength to steel of the present invention when added in an amount of at least 0.0001%.

[0044] Sulfur is not an essential element but may be contained as an impurity in steel and from point of view of the present invention the Sulfur content is preferably as low as possible but 0.09% or less from the viewpoint of manufacturing cost. Further if higher Sulfur is present in steel it combines to form Sulfides especially with Manganese and reduces its beneficial impact on the present invention.

[0045] Phosphorus constituent of the Steel of the present invention is from 0% to 0.09%, Phosphorus reduces the spot weldability and the hot ductility, particularly due to its tendency to segregate at the grain boundaries or co-segregate with Manganese. For these reasons, its content is limited to 0.09% and preferably lower than 0.06%.

[0046] Nitrogen is limited to 0.09% to avoid ageing of material and to minimize the precipitation of Aluminum nitrides during solidification which are detrimental for mechanical properties of the steel.

[0047] Chromium is an optional element that may constitutes of steel from 0% to 1%. Chromium provide strength to the steel by solid solution strengthening and a minimum of 0.1% is required to impart the strength but when used above 1% impairs surface finish of steel. The preferred limit for the presence of Chromium is from 0.1% to 0.5%.

[0048] Molybdenum is an optional element that constitutes 0% to 0.4% of the Steel of the present invention; Molybdenum plays an effective role in improving hardenability and hardness, delays the appearance of Bainite hence promote the formation of Martensite, in particular when added in an amount of at least 0.001% or even of at least 0.002%. However, the addition of Molybdenum excessively increases the cost of the addition of alloy elements, so that for economic reasons its content is limited to 0.4%.

[0049] Niobium is present in the Steel of the present invention from 0% to 0.1% and suitable for forming carbo-nitrides to impart strength of the Steel of the present invention by precipitation hardening. Niobium will also impact the size of microstructural components through its precipitation as carbo-nitrides and by retarding the recrystallization during heating process. Thus, finer microstructure formed at the end of the holding temperature and as a consequence after the complete annealing will lead to the hardening of the product. However, Niobium content above 0.1% is not economically interesting as a saturation effect of its influence is observed this means that additional amount of Niobium does not result in any strength improvement of the product.

[0050] Vanadium is effective in enhancing the strength of steel by forming carbides or carbo-nitrides and the upper limit is 0.1% from economic points of view.

[0051] Nickel may be added as an optional element in an amount of 0% to 1% to increase the strength of the steel the present invention and to improve its toughness. A minimum of 0.01% is preferred to get such effects. However, when its content is above 1%, Nickel causes ductility deterioration.

[0052] Copper may be added as an optional element in an amount of 0% to 1% to increase the strength of the of Steel of the present invention and to improve its corrosion resistance. A minimum of 0.01% is preferred to get such effects. However, when its content is above 1%, it can degrade the surface aspects.

[0053] Calcium can be added to the steel of the present invention in an among from 0.001% to 0.01%. Calcium is added to steel of the present invention as an optional element especially during the inclusion treatment. Calcium contributes towards the refining of the Steel by binding the detrimental Sulfur content in globular form thereby retarding the harmful effect of Sulfur.

[0054] Other elements such as Sn, Pb or Sb can be added individually or in combination in the following proportions: $\text{Sn} \leq 0.1\%$, $\text{Pb} \leq 0.1\%$ and $\text{Sb} \leq 0.1\%$. Up to the maximum content levels indicated, these elements make it possible to refine the grain during solidification. The remainder of the composition of the steel consists of iron and inevitable impurities resulting from processing.

[0055] The microstructure of the martensitic steel sheet will now be described in details, all percentages being in area fraction.

[0056] Martensite constitutes at least 92% of the microstructure by area fraction. The martensite of the present invention can comprise both fresh and tempered martensite. However, fresh martensite is an optional microconstituent which is preferably limited in the steel at an amount of from 0% to 4%, preferably from 0 to 2% and even better equal to 0%. Fresh martensite may form during cooling after tempering. Tempered martensite is formed from the martensite which forms during the second step of cooling after annealing and particularly after below M_s temperature and more particularly from $M_s - 10^\circ \text{C}$. to 20°C . Such martensite is then tempered during the holding at a tempering temperature T_{temper} from 150°C . to 300°C . The martensite of the present invention imparts ductility and strength to such steel. Preferably, the content of martensite is from 93% to 99% and more preferably from 94% to 98%.

[0057] The cumulated amount of ferrite and bainite represents from 1% to 8% of the microstructure. The cumulative presence of bainite and ferrite does not affect adversely to the present invention till 8% but above 8% the mechanical properties may get impacted adversely. Hence the preferred limit for the cumulative presence ferrite and bainite is kept from 1% to 7% and more preferably from 1% to 6%.

[0058] Bainite forms during the reheating before tempering. In a preferred embodiment, the steel of the present invention contains 1 to 3% of bainite. Bainite can impart formability to the steel but when present in a too big amount, it may adversely impact the tensile strength of the steel.

[0059] Ferrite may form during the first step of cooling after annealing but is not required as a microstructural constituent. Ferrite formation must be kept as low as possible and preferably less than 2% or even less than 1%.

[0060] Residual Austenite is an optional microstructure that can be present from 0% to 2% in the steel.

[0061] In addition to the above-mentioned microstructure, the microstructure of the cold rolled martensitic steel sheet is free from microstructural components such as pearlite and cementite.

[0062] The steel according to the invention can be manufactured by any suitable methods. It is however preferable to use the method according to the invention that will be detailed, as a non-limitative example.

[0063] Such preferred method consists in providing a semi-finished casting of steel with a chemical composition of the prime steel according to the invention. The casting can be done either into ingots or continuously in form of thin slabs or thin strips, i.e. with a thickness ranging from approximately 220 mm for slabs up to several tens of millimeters for thin strip.

[0064] For example, a slab having the chemical composition according to the invention is manufactured by continuous casting wherein the slab optionally underwent a direct soft reduction during the continuous casting process to avoid central segregation and to ensure a ratio of local Carbon to nominal Carbon kept below 1.10. The slab provided by continuous casting process can be used directly at a high temperature after the continuous casting or may be first cooled to room temperature and then reheated for hot rolling.

[0065] The temperature of the slab, which is subjected to hot rolling, must be at least 1000°C . and must be below 1280°C . In case the temperature of the slab is lower than 1280°C ., excessive load is imposed on a rolling mill and, further, the temperature of the steel may decrease to a Ferrite transformation temperature during finishing rolling, whereby the steel will be rolled in a state in which transformed Ferrite contained in the structure. Therefore, the temperature of the slab must be high enough so that hot rolling should be completed in the temperature range of Ac_3 to $\text{Ac}_3 + 100^\circ \text{C}$. Reheating at temperatures above 1280°C . must be avoided because they are industrially expensive.

[0066] The sheet obtained in this manner is then cooled at a cooling rate of at least $20^\circ \text{C}/\text{s}$ to the coiling temperature which must be below 650°C . Preferably, the cooling rate will be less than or equal to $200^\circ \text{C}/\text{s}$.

[0067] The hot rolled steel sheet is then coiled at a coiling temperature below 650°C . to avoid ovalization and preferably from 475°C . to 625°C . to avoid scale formation, with an even preferred range for such coiling temperature from 500°C . to 625°C . The coiled hot rolled steel sheet is then cooled down to room temperature before subjecting it to optional hot band annealing.

[0068] The hot rolled steel sheet may be subjected to an optional scale removal step to remove the scale formed during the hot rolling before optional hot band annealing. The hot rolled sheet may then have subjected to an optional hot band annealing. In a preferred embodiment, such hot band annealing is performed at temperatures from 400°C . to 750°C ., preferably for at least 12 hours and not more than 96 hours, the temperature preferably remaining below 750°C . to avoid transforming partially the hot-rolled microstructure and, therefore, possibly losing the microstructure homogeneity. Thereafter, an optional scale removal step of this hot rolled steel sheet may be performed through, for example, pickling of such sheet.

[0069] The hot rolled steel sheet thus obtained preferably has a thickness from 0.5 mm to 10 mm and more preferably from 1 mm to 8 mm and even more preferably from 1 mm to 6 mm. This hot rolled steel sheet may optionally be send directly for the heat treatment to manufacture an optional Hot rolled heat treated martensitic steel suitable for yellow and green goods. The heat treatment process for both the Hot rolled steel sheet and cold rolled steel sheet are identical.

[0070] This hot rolled steel sheet is then subjected to cold rolling to obtain a cold rolled steel sheet with a thickness reduction from 35 to 90%.

[0071] The cold rolled steel sheet is then heated in a two steps heating process wherein the first step of heating starts from room temperature, the cold rolled steel sheet being heated, at a heating rate HR_1 of at least $10^\circ \text{C}/\text{s}$, to a temperature HT_1 which is in a range from 410°C . to 750°C . In a preferred embodiment, the heating rate HR_1 for such first step of heating is at least $12^\circ \text{C}/\text{s}$ and more preferably at least $13^\circ \text{C}/\text{s}$. The preferred HT_1 temperature for such first step is from 425°C . to 725°C . and more preferably from 435°C . to 690°C .

[0072] In the second step of heating, the cold rolled steel sheet is heated from HT_1 to an annealing temperature T_{soak} which is from Ac_3 to $\text{Ac}_3 + 100^\circ \text{C}$., preferably from $\text{Ac}_3 + 10^\circ \text{C}$. to $\text{Ac}_3 + 100^\circ \text{C}$., at a heating rate HR_2 which is from $0.5^\circ \text{C}/\text{s}$ to $50^\circ \text{C}/\text{s}$. In a preferred embodiment, the heating rate HR_2 for the second step of heating is from $0.7^\circ \text{C}/\text{s}$ to $25^\circ \text{C}/\text{s}$ and more preferably $0.8^\circ \text{C}/\text{s}$ and $20^\circ \text{C}/\text{s}$, wherein Ac_3 for the steel sheet is calculated by using the following formula:

[00001]
$$\text{Ac}_3 = 910 - 203[\text{C}]^{(1/2)} - 15.2[\text{Ni}] + 44.7[\text{Si}] + 104[\text{V}] + 31.5[\text{Mo}] + 13.1[\text{W}] - 30[\text{Mn}] - 11[\text{Cr}] - 20[\text{Cu}] + 700[\text{P}] + 400[\text{Al}] + 120[\text{As}] + 400[\text{Ti}]$$
wherein the elements contents are expressed in weight percentage of the cold rolled steel sheet.

[0073] The cold rolled steel sheet is held at T_{soak} during 10 seconds to 500 seconds to ensure a complete recrystallization and full transformation to austenite of the strongly work hardened initial structure.

[0074] The cold rolled steel sheet is then cooled in a two steps cooling process wherein the first step of cooling starts from T_{soak} , the cold rolled

steel sheet being cooled down, at a cooling rate CR1 from 15° C./s to 150° C./s, to a temperature T1 which is in a range from 620° C. to 750° C. In a preferred embodiment, the cooling rate CR1 for such first step of cooling is from 20° C./s to 120° C./s. The preferred T1 temperature for such first step is from 630° C. to 725° C.

[0075] In the second step of cooling, the cold rolled steel sheet is cooled down from T1 to a temperature T2 which is from Ms—10° C. to 20° C., at a cooling rate CR2 of at least 50° C./s. In a preferred embodiment, the cooling rate CR2 for the second step of cooling is at least 100° C./s and more preferably at least 150° C./s. The preferred T2 temperature for such second step is from Ms—50° C. to 20° C.

[0076] Ms for the steel sheet is calculated by using the following formula:

[00002]

$$Ms = 545 - 601 \cdot 2 * (1 - \exp(-0.868[C])) - 34.4[Mn] - 13.7[Si] - 9.2[Cr] - 17.3[Ni] - 15.4[Mo] + 10.8[V] + 4.7[Co] - 1.4[Al] - 16.3[Cu] - 361[Nb] -$$

[0077] Thereafter the cold rolled steel sheet is reheated to a tempering temperature Ttemper from 150° C. to 300° C. with a heating rate of at least 1° C./s and preferably of at least 2° C./s and more of at least 10° C./s during 100 s and 650 s. The preferred temperature range for tempering is from 200° C. to 300° C. and the preferred duration for holding at Ttemper is from 200 s to 600 s.

[0078] Then, the cold rolled steel sheet is cooled down to room temperature to obtain a cold rolled martensitic steel.

[0079] The cold rolled martensitic steel sheet of the present invention may optionally be coated with zinc or zinc alloys, or with aluminum or aluminum alloys to improve its corrosion resistance.

EXAMPLES

[0080] The following tests, examples, figurative exemplification and tables which are presented herein are non-restricting in nature and must be considered for purposes of illustration only and will display the advantageous features of the present invention.

[0081] Steel sheets made of steels with different compositions are gathered in Table 1 wherein the each element is denoted by its presence in weight percentage and the remaining is Iron and other process impurities, where the steel sheets are produced according to process parameters as stipulated in Table 2, respectively. Thereafter Table 3 gathers the microstructures of the steel sheets obtained during the trials and table 4 gathers the result of evaluations of obtained properties.

TABLE-US-00001 TABLE 1 Steels C Mn Si Al Ti S P N Cr Nb B A 0.10 2.11 0.22 0.03 0.017 0.0012 0.0020 0.0036 0.03 0.002 0.0028 B 0.10 2.10 0.23 0.02 0.023 0.0024 0.0018 0.0045 0.03 0.002 0.0041 C 0.10 2.11 0.22 0.03 0.017 0.0008 0.0019 0.0034 0.03 0.002 0.0028 D 0.10 2.09 0.22 0.03 0.018 0.0020 0.0017 0.0034 0.03 0.002 0.0032 E 0.11 2.18 0.50 0.04 0.015 0.0015 0.0012 0.0040 0.03 0.011 0.0004 F 0.11 2.14 0.50 0.04 0.016 0.0011 0.0011 0.0050 0.03 0.011 0.0003 G 0.10 2.16 0.46 0.05 0.014 0.0012 0.0011 0.0057 0.02 0.012 0.0004 H 0.15 1.88 0.20 0.03 0.023 0.0019 0.0018 0.0041 0.18 0.005 0.0001 according to the invention; R = reference; underlined values: not according to the invention.

Table 2

[0082] Table 2 gathers the hot rolling and annealing process parameters implemented on cold rolled steel sheets to impart the steels of table 1 with requisite mechanical properties to become a martensitic steel.

TABLE-US-00002 The table 2 is as follows: Hot rolling Cooling Step 2 rate to Annealing Step 1 Cooling Cooling Reheating FRT coiling Coiling CR HR1 HT1 HR2 Tsoak Annealing CR1 CR2 Trials Steel (° C.) (° C.) (° C./s) (° C.) (%) (° C./s) (° C.) (° C./s) (° C.) time(s) (° C./s) T1 (° C.) (° C./s) T2 (° C.) I1 A 1230 870 30 530 57 15 610 0.9 865 285 45 635 617 20 I2 B 1230 870 30 530 63 15 550 1.3 860 249 53 640 711 20 I3 C 1230 870 30 530 50 13 550 1.1 860 295 32 630 589 20 I4 D 1230 870 30 530 55 13 445 1.8 860 236 43 640 749 20 I5 E 1150 890 30 530 53 14 600 0.9 850 295 33 680 638 20 I6 F 1150 890 30 530 61 15 600 1.1 860 260 43 635 675 20 R1 G 1150 890 30 530 63 12 450 1.5 800 236 47 615 719 20 R2 H 1150 895 30 530 50 15 550 1.1 880 290 34 600 538 20 Tempering Heating rate to Ttemper Holding Ms Ac3 Trials Steel Ttemper (° C./s) (° C.) time (s) (° C.) (° C.) I1 A 12 225 261 405 821 I2 B 14 230 230 404 823 I3 C 12 230 271 405 821 I4 D 15 230 217 407 824 I5 E 12 230 217 403 829 I6 F 13 230 238 404 830 R1 G 12 200 217 404 831 R2 H 12 230 290 403 816 I = according to the invention; R = reference; underlined values: not according to the invention.

[0083] Table 3 exemplifies the results of the tests conducted in accordance with the standards on different microscopes such as Scanning Electron Microscope for determining the microstructures of both the inventive and reference steels in terms of area fraction. The results are stipulated herein: TABLE-US-00003 TABLE 3 Tempered Fresh Ferrite + Residual Trials Steel Martensite Martensite Bainite Austenite I1 A 98 0 2 0 I2 B 98 0 2 0 I3 C 98 0 2 0 I4 D 98 0 2 0 I5 E 97 0 3 0 I6 F 95 0 5 0 R1 G 60 0 40 0 R2 H 89 0 11 0 I = according to the invention; R = reference; underlined values: not according to the invention.

Table 4

[0084] The results of the various mechanical tests conducted in accordance to the standards are gathered. The ultimate tensile strength, yield strength and total elongation are tested in accordance of ISO-6892. To estimate hole expansion, a test called hole expansion is applied, in this test sample is subjected to punch a hole of 10 mm and deformed after deformation we measure the hole diameter and calculate $HER\% = 100 * (Df - Di) / Di$

TABLE-US-00004 Tensile Tensile Yield Strength Strength Strength Yield Total Total (MPa)- (MPa)- (MPa)- Strength Elongation Elongation HER Trials Steels RD TD RD (MPa)-TD (%) RD (%) TD (%) I1 A 1242 1253 1109 1102 4.6 3.7 46.8 I2 B 1232 1248 1137 1088 4.9 4.0 65.0 I3 C 1232 1256 1132 1099 4.7 4.7 52.5 I4 D 1240 1256 1103 1113 6.0 4.7 54.8 I5 E 1267 1293 1156 1119 6.2 5.3 81.2 I6 F 1221 1241 1106 1044 6.1 5.4 54.3 R1 G 1000 1017 687 661 12.6 10.9 38.0 R2 H 1190 1207 983 932 7.6 7.0 21.3 I = according to the invention; R = reference; underlined values: not according to the invention.

Claims

1-15. (canceled)

16: A martensitic steel sheet comprising the following composition, expressed in percentage by weight: $0.08\% \leq C \leq 0.14\%$; $1.95\% \leq Mn \leq 2.6\%$; $0.1\% \leq Si \leq 0.8\%$; $0.01\% \leq Al \leq 0.1\%$; $0.001\% \leq Ti \leq 0.1\%$; $0.0001\% \leq B \leq 0.05\%$; $0\% \leq S \leq 0.09\%$; $0\% \leq P \leq 0.09\%$; $0\% \leq N \leq 0.09\%$; and can contain one or more of the following optional elements $0.1\% \leq Cr \leq 1\%$; $0\% \leq Ni \leq 1\%$; $0\% \leq Cu \leq 1\%$; $0\% \leq Mo \leq 0.4\%$; $0\% \leq Nb \leq 0.1\%$; $0\% \leq V \leq 0.1\%$; $0\% \leq Sn \leq 0.1\%$; $0\% \leq Pb \leq 0.1\%$; $0\% \leq Sb \leq 0.1\%$; $0.001\% \leq Ca \leq 0.01\%$; a remainder of the composition being composed of iron and unavoidable impurities caused by processing, a microstructure of the steel comprising, by area percentage, at least 92% of martensite, a cumulated amount of ferrite and bainite from 1% to 8%, and an optional amount of residual austenite from 0% to 2%.

17: The martensitic steel sheet according to claim 16 wherein the composition includes 0.15% to 0.7% of Silicon.

18: The martensitic steel sheet according to claim 16 wherein the composition includes 0.09% to 0.13% of Carbon.

19: The martensitic steel sheet according to claim 16 wherein the composition includes 2% to 2.5% of Manganese.

20: The martensitic steel sheet according to claim 16 wherein the amount of martensite is from 93% to 99%.

21: The martensitic steel sheet according to claim 16 wherein the sheet has a hole expansion ration of more than 40%.

22: A method of production of a martensitic steel sheet comprising the following successive steps: providing a semi-finished product with a composition, expressed in percentage by weight: $0.08\% \leq C \leq 0.14\%$; $1.95\% \leq Mn \leq 2.6\%$; $0.1\% \leq Si \leq 0.8\%$; $0.01\% \leq Al \leq 0.1\%$; $0.001\% \leq Ti \leq 0.1\%$; $0.0001\% \leq B \leq 0.05\%$; $0\% \leq S \leq 0.09\%$; $0\% \leq P \leq 0.09\%$; $0\% \leq N \leq 0.09\%$; and can contain one or more of the following optional elements $0.1\% \leq Cr \leq 1\%$; $0\% \leq Ni \leq 1\%$; $0\% \leq Cu \leq 1\%$; $0\% \leq Mo \leq 0.4\%$; $0\% \leq Nb \leq 0.1\%$; $0\% \leq V \leq 0.1\%$; $0\% \leq Sn \leq 0.1\%$; $0\% \leq Pb \leq 0.1\%$; $0\% \leq Sb \leq 0.1\%$; $0.001\% \leq Ca \leq 0.01\%$; a remainder of the composition being composed of iron and unavoidable impurities caused by processing; reheating the semi-finished product to a temperature from 1000° C. to 1280° C.; rolling the semi-finished product in the austenitic range wherein the hot rolling finishing temperature is from Ac3 to Ac3+100° C. to obtain a hot rolled steel sheet; cooling the sheet at a cooling rate of at least 20° C./s to a coiling temperature below 650° C.;

and coiling the hot rolled sheet; cooling the hot rolled sheet to room temperature; optionally performing scale removal process on the hot rolled steel sheet; optionally annealing may be performed on hot rolled steel sheet; optionally performing scale removal process on the hot rolled steel sheet; then cold rolling the hot rolled steel sheet with a reduction rate CR from 35 to 90% to obtain a cold rolled steel sheet; then heating the cold rolled steel sheet with a first heating step and a second heating step wherein: the first heating step starts from room temperature to a temperature HT1 from 410° C. to 750° C., with a heating rate HR1 of at least 10° C./s; the second heating step starts from HT1 to a temperature Tsoak from Ac3 to Ac3+100° C., with a heating rate HR2 from 0.5° C./s to 50° C./s, and holding for 10 to 500 seconds; then cooling the cold rolled steel sheet with a first cooling step and a second cooling step wherein: the first cooling step starts from Tsoak down to a temperature T1 from 620° C. to 750° C., with a cooling rate CR1 from 15° C./s to 150° C./s; the second cooling step starts from T1 down to a temperature T2 from Ms-10° C. to 20° C., with a cooling rate CR2 of at least 50° C./s, then reheating the cold rolled steel sheet at a rate of at least 1° C./s to a tempering temperature Ttemper from 150° C. to 300° C. and holding for 100 to 650 seconds; then cooling to room temperature with a cooling rate of at least 1° C./s to obtain a martensitic steel sheet.

23: The method according to claim 22 wherein the coiling temperature is from 475° C. to 625° C.

24: The method according to claim 22 wherein Tsoak is from Ac3+10° C. to Ac3+100° C.

25: The method according to claim 22 wherein CR1 is from 20° C./s to 120° C./s.

26: The method according to claim 22 wherein T1 is from 630° C. to 725° C.

27: The method according to claim 22 wherein CR2 is greater than 100° C./s.

28: The method according to claim 22 wherein T2 is from Ms—50° C. and 20° C.

29: The method according to claim 22 wherein Ttemper is from 200° C. to 300° C.

30: A method for manufacturing a structural part of a vehicle comprising employing the steel sheet as recited in claim 16.

31: A method for manufacturing a structural part of a vehicle comprising employing the steel sheet manufactured according to the method as recited in claim 22.
